Synchronization

Introduction to Synchronization

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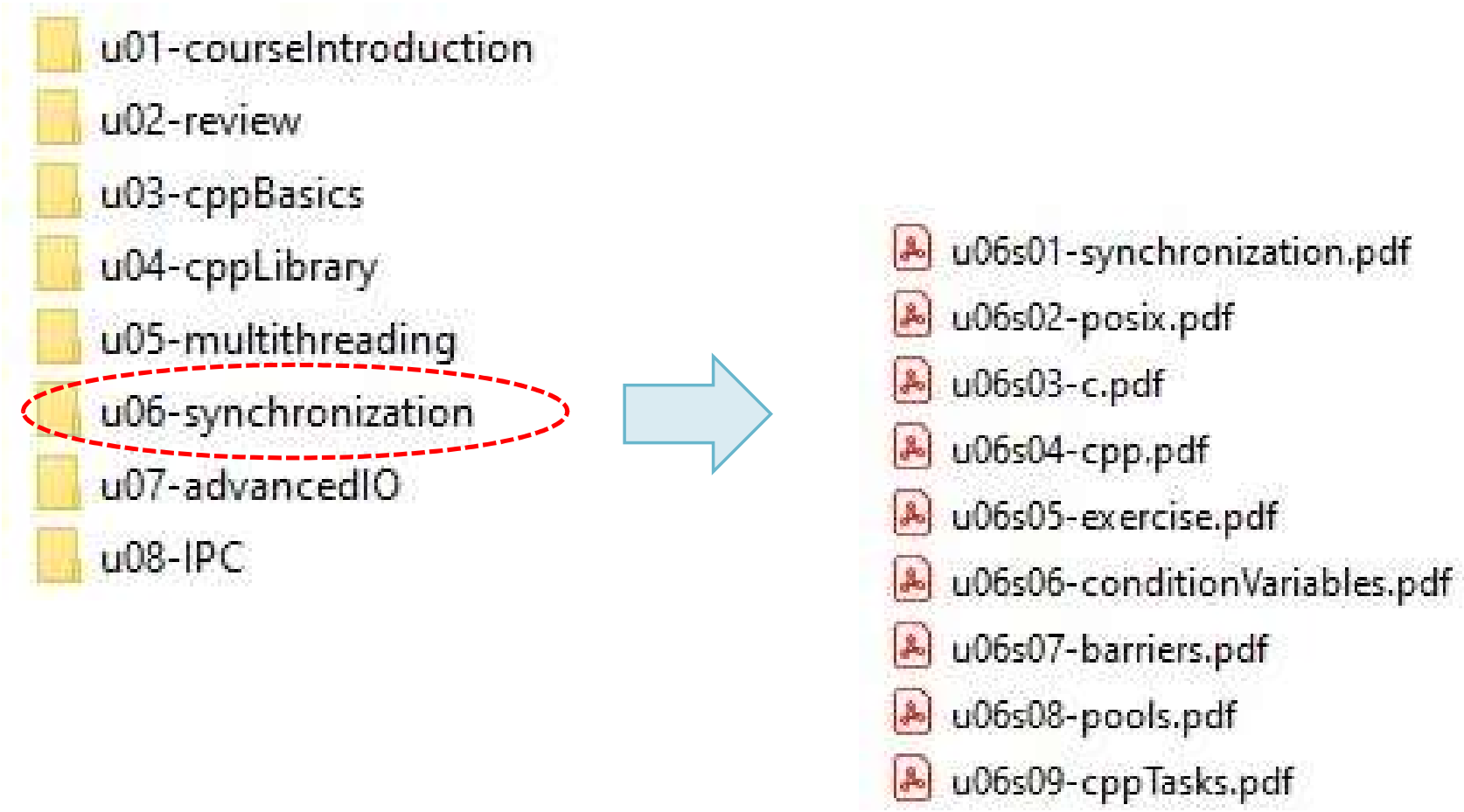
Introduction

- Where are we?



Introduction

- Critical Section (**CS**) or Critical Region (**CR**)
  - A section of code, common to multiple threads, in which each thread can read and **write** shared objects
- Access to CS is subject to **race conditions**
  - The result depends on the execution order of the processes instructions

Example

FIFO, Queue, Circular Buffer

T_i

```
void enqueue (int val) {
  if (n>SIZE) return;
  queue[tail] = val;
  tail=(tail+1)%SIZE;
  n++;
  return;
}
```

register = n
register = register + 1
n = register

T_j

```
int dequeue (int *val) {
  if (n<=0) return;
  *val=queue[head];
  head=(head+1)%SIZE;
  n--;
  return;
}
```

register = n
register = register - 1
n = register

- Even if **enqueue** and **dequeue** operate on the different ends of the queue, the variable **n** is shared

Race condition:
Increments and decrements can be lost

Critical sections

- Race conditions could be prevented with an access protocol that enforces mutual exclusion for each CS
  - Before a CS, there should be a **reservation** section
    - The reservation code must block (lock out) the P (or T) if another P (or T) is using its CS
  - After the CS, there should be a **release** section
    - The release possibly unlocks another P (or T) which was waiting in the ”reservation” code of its CS

Access protocol

T_i

```
while (TRUE) {
   ...
   reservation code
   Critical Section
   release code
   ...
   non critical section
}
```

T_j

```
while (TRUE) {
   ...
   reservation code
   Critical Section
   release code
   ...
   non critical section
}
```

- Every Critical Section is protected by an
  - Enter code (reservation, or prologue)
  - Exit code (release, or epilogue)
- Non-critical sections should not be protected
- OSs provide appropriate primitives